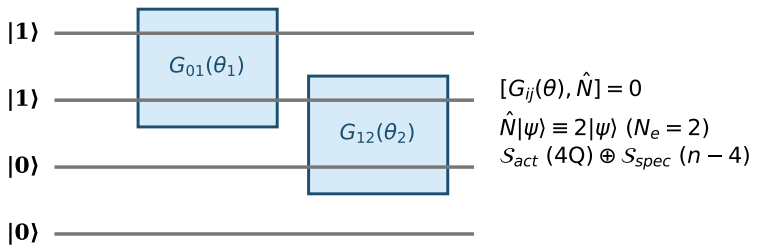
(a) Target Algorithmic Workloads & Symmetries

Fermionic VQE: Excitation-Conserving Givens Network



$[G_{ij}(\theta), \hat{N}] = 0$
$\hat{N}|\psi\rangle \equiv 2|\psi\rangle$ $(N_e = 2)$
$\mathcal{S}_{act}$ (4Q) $\oplus$ $\mathcal{S}_{spec}$ $(n-4)$

QAOA Max-Cut: Parity-Conserving Alternating Ansatz

- Unitary: $|\psi_p\rangle = \prod_{l=1}^{p} e^{-i\beta_l H_B} e^{-i\gamma_l H_C} |+\rangle^{\otimes n}$
- Exact Parity: $[\hat{\mathcal{P}}, H_B] = [\hat{\mathcal{P}}, H_C] = 0 \Rightarrow \hat{\mathcal{P}}|\psi_p\rangle \equiv +1|\psi_p\rangle$

Algebraic Symmetries

(b) Exact Algorithmic Invariants (O(1) Scaling)

1. Particle-Number Sector Invariant (VQE):

$\langle \hat{N} \rangle \equiv N_e = 2.0000, \quad \mathrm{Var}(\hat{N}) \equiv 0.0000, \quad \mathcal{L}_{leak} \equiv 0.0000$

2. Global $\mathbb{Z}_2$ Parity Invariant (QAOA):

$\langle \hat{\mathcal{P}} \rangle = \langle \bigotimes_{j=1}^{n} X_j \rangle \equiv +1.0000$

(Holds for all depths $p \geq 1$, angles $\vec{\gamma}, \vec{\beta}$, and graphs G)

3. Unitary Retrace Echo & Zero-Simulation Scaling:

$F_{echo} \equiv 1.0000 \Longrightarrow$ **O(1)** Ref. / **O(S⌈n/64⌉)** POPCNT ($n = 4$–64)

Exact Invariants (O(1))

(c) Heavy-Hex Calibration Mapping

Greedy Subgraph Optimization on 156-Qubit Heron r2:

$C^* = \arg\min_{C \subset \mathcal{G}_{hw}} [\sum_k \varepsilon_{CZ}(q_k, q_{k+1}) + 0.5 \sum_k \varepsilon_{meas}(q_k)]$

Live t_{submit}/t_{exec} calibration queries for chains $n \in \{4, \ldots, 64\}$.

Target Circuits

kingston: 50 — 49 — 48 — 47 $\quad \bar{\varepsilon}_{CZ} = 0.19\%, \; T_1 = 272\ \mu s$

fez: 109 — 110 — 111 — 98 $\quad \bar{\varepsilon}_{CZ} = 0.21\%, \; T_1 = 147\ \mu s$

**Baseline ($n = 4$) chains shown; Campaign 2 routes $n \in \{4$–$64\}$ (Tables III & V).*

Batched ISA Payload

(d) Dual 156-Qubit IBM Heron r2 H/W Execution

Single-Job Batched Execution (SamplerV2, 8,192 shots/circuit):

- Dual processors: ibm_kingston & ibm_fez (Heron r2)
- Campaign 1 ($n = 4$ baseline): **13.0 s**/QPU (26.0 s total, 81,920 shots)
- Campaign 2 ($n = 4$–64 scaling): **69.0 s** & **68.0 s** (137.0 s, 491,520 shots)
- Cumulative Total: **163.0 s** QPU time across 70 circuits (**573,440** shots)

Dual Single-Basis Readout Pipelines:

- Z-Basis Readout: Computational basis for Hamming weights $w(x) = \sum x_j \Rightarrow P(w)$ population statistics
- X-Basis Readout: Terminal $H^{\otimes n}$ rotation for parity $\langle \hat{\mathcal{P}} \rangle = \sum_y [C(y)/S](-1)^{\sum y_j}$

QPU Counts (573,440 Shots)

(e) Simulation-Free Invariant Diagnostic Engine & Error-Sensitive Indicators

1. Coherent Over-Rotation

- Echo F_{echo} ($n = 4$ | 64Q act):
  Kingston: 80.83% | 78.78%
  Fez: 80.13% | 70.95%
- Physical Diagnostic:
  Angle errors $\theta \to \theta + \delta\theta$ strictly preserve $[G, \hat{N}] = 0$ ($\mathcal{L}_{leak} \equiv 0$), but cause retrace decay $U^\dagger U \neq I$.

2. Asymmetry & SPAM Filter

- In-Sector $P(w = 2)$ ($n = 4$ | 64Q):
  Kingston: 91.63% | 87.29% (act)
  Fez: 88.39% | 82.85% (act)
- Asymmetry & Deconvolution:
  $R_{asym} = P(w = 1)/P(w = 3) =$ **3.00**
  Spectator filter $\hat{\varepsilon}_{0 \to 1}$ isolates T_1 decay from readout dark counts.

3. Dephasing & Marginals

- Global Parity ($n = 4, \; p = 1 \to 2$):
  Kingston: $+0.826 \to +0.733$
  Fez: $+0.785 \to +0.686$
- 64Q Marginal Retrace ($k = 1$):
  Kingston: 55.54% | Fez: 58.84%
  Phase flips $\{Z_u, \hat{\mathcal{P}}\} = 0$ drive depth decay $\langle \hat{\mathcal{P}} \rangle \approx (1 - 2\varepsilon_\phi)^n$.